SPACE AND TIME

A concise introduction to space, time and spacetime

1. Meaning of Space

Space is the three-dimensional extent in which objects exist and events occur. It describes the positions of objects and the distances between them.

The three dimensions of space are:

- Length
- Width
- Height

Properties of space:

- Space has three dimensions.
- Objects occupy space.
- Space contains matter and energy.
- Distance is commonly measured in metres (m).

2. Meaning of Time

Time is the continuous progression of events from the past, through the present, into the future. It helps us describe the duration and sequence of events.

Properties of time:

- Time is measured using clocks.
- The SI unit of time is the second (s).
- Time can also be expressed in minutes and hours.
- Time is used to describe motion and change.

3. Relationship Between Space and Time

Space and time are closely connected. In Einstein's theory of relativity, they are treated together as a four-dimensional structure called spacetime: three dimensions of space and one dimension of time.

Mass and energy affect the geometry of spacetime. In general relativity, massive objects such as stars and planets curve spacetime, and this curvature is associated with the gravitational effects we observe.

4. Important Physics Formulas

Quantity	Formula	Meaning
Speed	$v = d \div t$	Distance travelled per unit time
Distance	$d = v \times t$	Speed multiplied by time
Time	$t = d \div v$	Distance divided by speed
Acceleration	$a = (v - u) \div t$	Change in velocity per unit time

Symbols: v = speed (or final velocity where specified), d = distance, t = time, a = acceleration, and u = initial velocity. In the acceleration formula, v represents final velocity.

5. Space-Time in Relativity

Albert Einstein's theory of relativity explains how measurements of space and time depend on an observer's motion and, in general relativity, on gravity.

Two important ideas are:

1. **Time dilation:** Time can pass at different rates for observers moving relative to one another, or located in regions with different gravitational strengths.
2. **Length contraction:** An object moving at a speed close to the speed of light is measured to be shorter along its direction of motion by an observer relative to whom it is moving.

The speed of light in a vacuum is approximately:

$$\mathbf{c = 3.0 \times 10^8\ m/s}$$

6. Difference Between Space and Time

SPACE	TIME
Describes where an event happens.	Describes when an event happens.
Has three spatial dimensions.	Is the fourth dimension in spacetime.
SI unit of distance: metre (m).	SI unit: second (s).
Describes position and distance.	Describes duration and sequence.

7. Applications of the Study of Space and Time

- Satellite communication
- GPS navigation
- Astronomy and space exploration
- Understanding gravity
- Studying the motion of planets
- Understanding the structure and history of the universe

8. Summary

Space describes the location and distance of objects, while time describes the duration and order of events. Modern physics combines them into spacetime, a central concept in Einstein's theory of relativity.

SPACE = WHERE • TIME = WHEN • SPACETIME = THEIR CONNECTION

9. Revision Questions

3. Define space.
4. What is time?
5. State the three dimensions of space.
6. What is spacetime?
7. Who developed the theory of relativity?
8. State the SI units of distance and time.
9. What is time dilation?
10. State two applications of the study of spacetime.